Course Syllabus

MATH 107: Introduction to Linear Algebra

MATH 107: Introduction to Linear Algebra

Cal Poly Humboldt — Fall 2026

Course Information

Course Number: MATH 107
Course Title: Introduction to Linear Algebra
Class Number: 42493
Section: 1
Session: Regular Academic Session
Units: 3
Semester: Fall 2026
Mode of Instruction: Face to Face Instruction ONLY
General Education: GE Area 2 — Mathematics

Lecture

Days/Time: Monday, Wednesday, Friday, 10:00–10:50 a.m.
Location: FR 201

Final Exam

Wednesday, December 16, 2026, 10:20 a.m.–12:10 p.m., FR 201. The final exam is required.

Instructor Information

Teacher: Dr. Rosanna Overholser
Email: rho3@humboldt.edu
Office Location: BSS 334
Office Phone: 707-826-4022
Office Hours: TBD (or by appointment)

Please use your Cal Poly Humboldt email account to contact me and check both Canvas and your Cal Poly Humboldt email regularly for course communications.

Course Description (Catalog Description)

Euclidean spaces, matrices and matrix arithmetic, solving systems of equations, and eigenvalues and eigenvectors. Emphasis on applications and use of computation.

Division: Lower Division. **Weekly:** Lecture 3 hrs. **Instruction Mode:** Face to face only.

Prerequisite: (MATH 101 and MATH 101T) or MATH 102 or ALEKS PPL assessment score 78–100 or appropriate high school coursework. All prerequisite courses must be passed with a grade of C− or higher. See Math Placement and ALEKS PPL for details.

Note: if you plan to use MATH 107 as a prerequisite for another MATH, DATA, or STAT course, you must have a grade of C− or better in this course.

Course Learning Outcomes

By the end of this course, students will be able to:

1. Interpret matrices, vectors, and complex numbers, both algebraically and geometrically.
2. Solve systems of linear equations using row-reduction techniques by hand and with software.
3. Find eigenvalues and eigenvectors of a matrix by hand for a 2×2 matrix and with software for general cases.
4. Understand and apply eigenvalues and eigenvectors to solve problems.
5. Recognize special types of matrices and their properties, such as triangular, symmetric, and positive definite matrices.
6. Communicate in the language of linear algebra, including linear combination, subspace, span, independence, basis, and dimension.
7. Write or adapt programs to implement linear algebra concepts, such as matrix arithmetic, vectors, and plotting.
8. Apply linear algebra to real-world problems.
9. Learn independently as well as collaboratively. Develop study techniques that include pre- and post-reading of class materials and be able to critically analyze and implement solutions in a team setting.

Daily learning objectives for each meeting are on Lectures.

General Education & Institutional Learning Outcomes

This course fulfills **Lower Division General Education Area 2 (Mathematics)** and supports Cal Poly Humboldt's Institutional Learning Outcomes related to quantitative reasoning, critical thinking, and communication.

Course Materials & Workload

MATH 107 includes three weekly lecture meetings. Some Fridays (and a few other days) include a lab, quiz, or exam as marked on the schedule. There is no separate laboratory enrollment.

In this 3-unit course, students should expect to spend approximately **9 hours per week** on course-related work, including time in class (3 hours) and time outside of class on reading, practice, labs, and exam preparation (6 hours).

Required Materials

Course website: Lecture outlines, labs, and assessments are on this site. Daily reading is listed with each meeting on Lectures.
Textbooks: Stephen Boyd and Lieven Vandenberghe, *Introduction to Applied Linear Algebra: Vectors, Matrices, and Least Squares* (VMLS), free to read and download.
Gilbert Strang, *Introduction to Linear Algebra*, selected sections only (especially §3.5 and 6.1), as marked on Lectures.
Software: Octave (I provide an optional Python version). See Resources.
Other Technology: A scientific calculator for in-class assessments and regular access to a computer with reliable internet. Bring a laptop on lab days, or let me know a day ahead that I should bring one for you to use.
Writing tools: Bring paper and a pen or pencil to class.

Access & Affordability

Students who do not have personal access to required technology may use computers either in campus labs or checked out from the library. Calculators are also available for check out from the library's front desk. If you want to get a free used computer, please check to see if our campus's Reusable Office Supply Exchange (ROSE) has one available.

Fees

There are no additional course fees beyond standard tuition and registration fees.

Course Feedback & Assessment

Students will complete a variety of assessments (dates on the schedule and Assessments):

- Class participation (daily in-class activities)
- Labs (about every two weeks)
- In-class quizzes (ten quizzes, first 10 minutes of class; the lowest two scores are dropped)
- Two midterms (Friday, October 2 and Friday, November 13, in class)
- A required final exam (Wednesday, December 16, 10:20 a.m.–12:10 p.m., FR 201)
- A capstone project (end of the semester)

Grading

Component	Weight
Participation / in-class work	5%
Labs	10%
Quizzes	25%
Two midterms	35%
Capstone	10%
Final	15%
Total	**100%**

Final letter grades will be assigned based on overall performance according to the following:

A: 93–100% A−: 90–92% B+: 87–89% B: 83–86% B−: 80–82% C+: 77–79% C: 73–76% C−: 70–72% D: 60–69% F: 0–59%

Note:

- **Grade Mode:** Letter grade by default. Students may elect Credit/No Credit grading according to university deadlines and procedures.
- A minimum grade of **C−** is required for the course to count as a prerequisite for other MATH or DATA classes.
- If you stop attending class and stop doing assignments without communicating with me, you will receive a grade of WU (Withdrawal Unsatisfactory). Please communicate with me early if you are having difficulties: it's possible that I can help you get back on track, or help you obtain a more appropriate grade (W or I) for your situation.

Attendance & Participation

Regular attendance and active participation are expected and essential for success in this course.

If you must miss class, you are responsible for obtaining notes and keeping up with course material. Materials stay on this website; Canvas is used for announcements, lab and project submissions, and grades.

Late or Missed Work Policies

The policies for late or missed work are as follows:

- **Lecture attendance** is expected, but I know that sometimes you will need to miss class due to illness or other emergencies. Each student may miss up to 10 lecture days without penalty and without contacting me. Reserve these days for emergencies. In my experience, it is rare for a student to succeed if they regularly miss class or are absent for more than two weeks, so if you find yourself in that situation, please communicate with me and your advisor so we can help you find a solution.
- **Quizzes** are in the first 10 minutes of class. The lowest two quiz scores are dropped. If you must miss a quiz due to illness or another emergency, contact me to arrange a makeup.
- **Labs** are due on the dates on the schedule. After a week, late labs are not accepted.
- **Capstone** work is due on the dates on the schedule. Late project work cannot be turned in due to the end of the semester and course grades being due. If you are unable to complete the project in time, you should consider whether your situation is such that an Incomplete grade is appropriate. This would give you more time to complete the course.
- **Midterms and the final exam** are required. If you need to miss a midterm due to illness or another emergency, communicate with me as soon as possible to reschedule.

Note: Technology issues (internet access, device failure, software problems) are not automatically considered valid excuses for late or missed work; students are expected to plan accordingly.

AI Policy

Much of your course grade will be based on work you produce in class (quizzes and exams) without AI as a resource.

You may use AI, such as NotebookLM or ChatGPT, to help you learn the material or to get unstuck on labs and the capstone done outside of class.

For work that you do at home that is turned in for a grade I have two requirements:

1. If you used AI, acknowledge this briefly in the document (you don't need to state in detail how you used AI or give your prompts, just the name of the AI service you used).

2. Be responsible for the correctness and appropriateness of all (AI-assisted or not) materials you turn in.

I may use AI to assist me in my delivery of the course material, such as asking whether an assignment is appropriate for a college-level course or whether the wording of a quiz question could be improved. I will not use AI to grade your work or use an AI detection tool to check your work.

Course Topics & Schedule

Topics (Approximate Order)

- Vectors as representations; arithmetic, linear combinations, and inner products
- Linear and affine functions; norms, angle, neighbors, and clustering
- Linear dependence, span, and bases
- Matrices, Ax, transformations, and linear systems
- Dynamics, simulation, products, powers, and computational cost
- Inverses, rank, and determinants
- Eigenvectors and eigenvalues
- Least squares, fitting, and SVD
- Capstone: accuracy, memory, runtime, and communication

Summary of Schedule and Important Dates

Note: The schedule has a detailed list of the course. Please check it regularly for updates.

Date	Event
August 24	First class
August 28	Lab 0
September 4	Quiz 1
September 7	Labor Day (no class)
October 2	Midterm 1
November 11	Veterans Day (no class)
November 13	Midterm 2
November 23–27	Fall break (no class)
December 9	Capstone gallery
December 11	Final review (last class meeting)
December 16	Final exam, 10:20 a.m.–12:10 p.m., FR 201

Add/drop and other university deadlines follow the Cal Poly Humboldt academic calendar.

University Policies & Resources

Students are responsible for reviewing the Cal Poly Humboldt Syllabus Addendum, which includes policies and resources related to:

- Academic honesty
- Disability accommodations
- Emergency procedures
- Title IX
- Student conduct and support services

Accommodations

Students who wish to request disability-related accommodations should contact the Student Disability Resource Center (SDRC) as early as possible and then email me (Rosanna Overholser, rho3@humboldt.edu) your letter of accommodation.

Classroom & Online Conduct

Standards of respectful and professional behavior apply to all course interactions, including in-person classes, assignments, and email communication.

Changes to the Syllabus

This syllabus is subject to change. Any changes will be communicated through **Canvas announcements**.

License